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Design and realization of cutting simulation function of digital twin system of CNC machine tool

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Abstract

CNC machine tool processing is an important part of the current development of machinery modernization. In order to ensure the accuracy and stable quality of numerical control machining, the encoding of numerical control program needs to be a validation to avoid in the process of the program which may leads to under-cut issue or collision case. Therefore, in the numerical control machining application process, you need to use digital twin technology to numerical control machining simulation cutting simulation, to verify the numerical control process is reasonable with the accuracy of a comprehensive, objective judgment, to avoid various abnormal situations have occurred in the process and to improve processing and production efficiency and quality.

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Keywords: CNC machine tool; digital twin; cutting simulation ;

1. Introduction

With the rapid development of our country's industrialization, CNC machining technology has also been widely used in the country. Compared to the general machine tools, CNC machine tools have a lot of characteristic such as high precision, high degree of automation, high reliability, stable processing quality, reliable, high productivity and so on. In the early stages of design and production preparation machine, the real numerical machining debugging needs long time, and it is high risk and high cost. At the same time, the requirements of technicians are also have higher

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technical level requirements for CNC machine tool technicians. Therefore, in the actual production process, technical personnel's mistakes in the design of the CNC machine tool process will cause serious processing accidents.

The concept of digital twin refers to that through the data of the physical device, the virtual entity representing the physical device can be constructed in the virtual space, and this connection is not one-way and static, but is connected throughout the life cycle of the product¹. The digital twin is an entire team of objects in the physical world, and an identical entity in the digital world is constructed through digital mode.

In combination with CNC machine tools and digital twin technology can help reduce machining accidents and errors while simulating the machining results of the machine tool reasonably. At the same time, simulating the machining results of the machine tool can reduce production costs and trial cutting time. To a certain extent, shorten the development cycle and production preparation cycle of the machine tool, can improve its rapid response to the market and control of the production preparation cycle.

2. Technical background

2.1. CNC machine tools

The CNC machine tool is an automated machine tool equipped with a program control system. The control system can logically process a program with a control code or other symbolic instructions and decode it, express it with coded numbers, and input it into the numerical control device through an information carrier. After arithmetic processing, the numerical control device sends out various control signals to control the action of the machine tool, and automatically process the parts according to the shape and size required by the drawing.

The CNC machine tool can solve the problem in complex, small batch, precise and multi-variety parts processing way. It is a flexible and high-performance automatic machine tool, which represents the development direction of modern machine tool control technology, and is a typical electro-mechanical integration products.

2.2. Digital Twins

The concept of digital twin was first appeared in 2003 year, professor Grieves made a lifecycle management courses at the University of Michigan². Later, the defense of U.S. Department introduced the concept of digital twins into the health maintenance of spacecraft³, and defined it as a simulation process that integrates multiple physical quantities, multiple scales, and multiple probabilities, and builds its complete mapping based on the physical model of the aircraft. The virtual model using historical data and real-time updates of sensor data, life cycle process characterization and reflect the physical object⁴. Digital twin has the following characteristics⁵: 1. The integration of various data of physical objects is a faithful mapping of physical objects; 2. The full life cycle of a physical object, with which it coevolves and accumulates knowledge; 3. Not only can describe the physical object, and can optimize the physical object based on the model. The current digital twin of management concept has been applied and validated in some areas.

2.3. MQTT protocol

The MQTT protocol is a lightweight, message queue-based protocol that implements the distribution of communication messages through topics.⁶ The protocol is designed on top of the TCP protocol, through the proxy server to achieve message publishing and subscription.⁷ It provides one-to-many message distribution, and one publisher can correspond to multiple subscribers. The message broker appears as a server, and is responsible for receiving messages published by publishers and delivering messages to subscribers. News publishers and subscribers usually appears as a client, the client may be things device, also can be mobile applications and so on.⁸

3. System Architecture

The digital twin system of CNC machine tool is mainly composed of CNC machine tool entity, data acquisition client, data server and system client. The system architecture is shown in Fig.1. This paragraph mainly explains the realization of cutting simulation function.

CNC machine as the source of the data information entity, MQTT client of workpiece and tool information collected by the sensor in real time, MQTT server receives a subscription MQTT packed data publishing client. The system client obtains the workpiece and tool information in real time by subscribing to the information of the MQTT server. After inputting the cutting programming information, the cutting simulation function is started. During the cutting process, the user can observe the cutting process in real time through the transparent machine shell. After the cutting is completed, the workpiece is displayed.

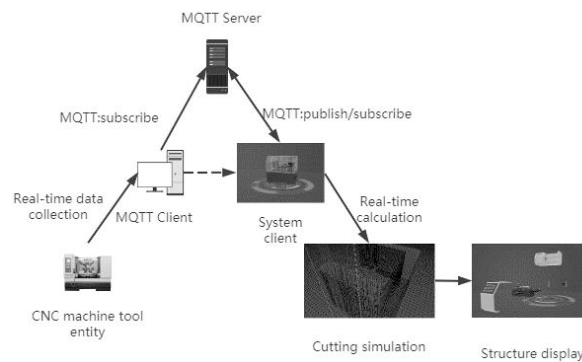


Fig.1. System architecture diagram

4. System implementation

4.1. The realization of communication based on MQTT protocol

- The application main process calls the system initialization module of the message proxy server to start the message proxy server function service. Whether the client is a message publisher or a message subscriber, it connects to the message proxy server through the client initialization module of the message proxy server.
- The system client (digital twin system of CNC machine tools) connects to the MQTT message broker server, completes registration to obtain a registered account, publishes / subscribes to its own topic, and the message broker server subscription table management module records the corresponding subscription topic and the account identification of the subscription client.
- The message publisher (numerical control machine entity) packages the data to be sent into a message and sends it to the MQTT message proxy server. The message proxy server stores it in the corresponding system receiving queue according to the message priority, and the message processing module is taken out of the queue. The message is pushed to the message receiving queue corresponding to the corresponding message subscriber according to the client subscription information recorded in the subscription table management module.
- The message listener of the MQTT message proxy server is triggered, and each message subscriber is notified to receive the message according to the information such as the message subscriber's account.
- The system client monitors a new message in the message queue and reads the message from the respective message receiving queue.

4.2. Workpiece simulation

Workpiece simulation is the core of the system, and how to perform efficient and accurate modeling is a problem that needs to be considered in the system development process. The workpiece simulation methods of CNC machine tools can be divided into four types, precision modeling methods, discrete vector methods, space division methods and hybrid modeling methods.

Precision modeling methods include boundary representation and constructed solid geometry method. The boundary representation can accurately express the entity and achieve accurate processing, but the continuous of Boolean operation of subtraction will greatly reduce the time and space performance. Constructed solid geometry method can also achieve accurate expression of entities. Boolean operations are suitable for expressing more regular-shaped geometry, but are not suitable for expressing free-form surfaces in processing. When the number of tool displacements is N , the simulation time complexity is $O(N^4)$.

Discrete vector method uses a series of discrete points and vectors with these points as the starting point to express the surface to be processed. Boolean subtraction is implemented by calculating the intersection of the tool scan volume and the vector to update the length of the vector. The method achieves high simulation efficiency, and it is easy to quantitatively calculate undercut or overcut through the measurement of vector length. The disadvantage of this method is that it is not suitable for situations where the direction of the surface vector changes dynamically and the entity cannot be directly constructed through simulation.

The space division method divides the workpiece space into a series of simple voxels. Among them, according to the different attributes of the workpiece space, it can be divided into object space method and image space method of workpiece. Advantage of the space division method is efficient and stable Boolean subtraction, idea of the method makes use divide and conquer processed Boolean operation between the solid model, to avoid the time-consuming three-dimensional surface intersecting line calculation. The disadvantage is that only voxels are used to express entities, which is an approximate entity expression method. When high-quality entity graphics rendering is required, the number of voxels in the entire entity model will be too large.

At this stage, hybrid modeling methods are used. This method combines the advantages of space division method and accurate modeling method. On the basis of preserving the efficient and stable Boolean subtraction operation, the unit voxel is added with triangular faces to be more accurate describes the contour surface of the entity.

4.3. Tool simulation

Tool scanning volume refers to the integration of the space occupied by the resulting entity by taking the tool as the scanning matrix and moving in a certain way along any path in space.⁹ The construction methods of the tool scan volume can be divided into two categories: explicit and implicit. Most of the explicit construction methods of the tool scan volume use orthogonal conditions to determine the intersection between the tool scan volume and the workpiece, and then construct the scan volume. The implicit construction method of the tool scan volume is to express the surface by establishing the relationship between the parameters. Compared with the explicit construction method, the implicit advantage is 1. The space point and the tool can be determined without error. The positional relationship of the scanning body, that is inside the scanning body, outside the scanning body and on the contour of the scanning body; 2. The method for determining the positional relationship is simple. Only by calculating the value of the implicit expression of the tool scan volume at the point to be measured, the position relationship can be determined, avoiding the time-consuming calculation of three-dimensional space intersection; 3. It is simple to solve the implicit equation of the tool scan volume. The tool scan body construction of this system adopts the implicit construction method.

4.4. Block Boolean operations

Block Boolean operations include intersection, union and subtraction operations of blocks. The block intersection operation refers to finding the common part of the subject and the object block and taking it as a new block; The block union operation refers to finding a body, which contains both the main block and the guest block; block Volume subtraction operation refers to removing the common part of the host and guest blocks from the main block,

and forming the remaining part into a new block. For the main block A and the guest block B, the formulas for Boolean intersection operation, union operation and subtraction operation can be expressed as

$$C_{\cap} = A \cap B \quad (1)$$

$$C_{\cup} = A \cup B \quad (2)$$

$$C_{-} = A - B \quad (3)$$

In the development process of this system, the block Boolean subtraction operation is mainly used for the cutting simulation between the workpiece and the tool.

4.5. Cutting simulation algorithm

The cutting simulation algorithm is mainly to update the workpiece model. Tool swept volume determined to be updated through the work area such as decompose or merge operation triangular sheet, and then the associated cam plate and tool scanning body intersection, to give the workpiece after machining Z the height value. This completes a cutting operation. Each workpiece update only considers the Z-map area related to the tool movement. For a certain Z-map area of a given workpiece model, if the area is within the range of the tool scan body, the workpiece model in this area is updated. Otherwise, not update the area. The update operation mainly involves the merging and decomposition of triangle pieces in the triangle binary tree LOD model. Among them, the merge is mainly realized by the edge collapse operation; the decomposition mainly uses the triangle binary tree to triangulate the area. After a series of triangle decomposition and merging operations, the updated workpiece model can better represent the processing situation of the free-form surface, and the different levels of detail can fully reflect the characteristics of the processed surface and can better approximate the original surface. Therefore, this algorithm can greatly increase the speed of graphics rendering on the basis of ensuring the simulation effect, thereby improving the quality and speed of simulation.

5. Cutting simulation function test

This experiment was implemented in the operating system of Windows 10, operating system version is 19383.1139, Unity 3D version is Unity 2017.4.14f1 (64-bit). The experimental content is three-axis numerical control machine tool processing simulation.

After the software is opened, enter the system interface. The system client obtains the workpiece and tool point cloud information by subscribing to the information of the MQTT server to generate the workpiece and tool model, and then starts the simulation cutting after setting the machine tool processing program. The operator can adjust the perspective (Fig.2) Observe the movement of the machine tool and the cutting of the workpiece in the scene. You can also observe the structure of the system through the exploded diagram (Fig.3) and the transparent view (Fig.4).

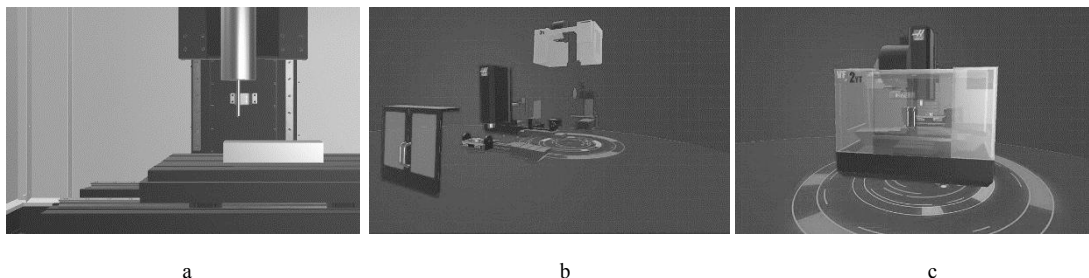


Fig. 2. (a)Cutting observation diagram;(b) Exploded view of structure;(c) Transparent machine diagram

6. Conclusion

In this article, the real-time cutting simulation function of the digital twin system of CNC machine tools is realized. This function can perform model cutting simulation prediction by scanning workpieces and tools. It can be applied to the trial cutting function of CNC machine tools and has practical application value.

The cutting simulation function of this system has the following advantages: 1. It simulates the cutting process of real materials, which can reduce the material waste during the machining process of the machine tool and reduce the material cost during the trial cutting process. 2. CNC machine tools are expensive. Simulation cutting through the digital twin system of CNC machine tools can ensure that CNC machine tools can be used for formal production and processing to the greatest extent, reducing collision accidents caused by program setting errors, reducing equipment maintenance costs, and improving machine tool equipment Life. 3. Using digital twin technology to simulate the cutting results and predict the real machining process, it has the characteristics of high efficiency and accuracy, and can reduce the time cost in the trial cutting process. The next step is to further improve the digital twin system and improve system performance .

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